HeartSense AI – Data Dictionary

1. User Table

Field Name	Data Type	Key	Description
User_id	INT	PK	Unique user identifier
Full_name	VARCHAR	—	User's full name
Email	VARCHAR	UNIQUE	User email address
Password	VARCHAR	—	Encrypted password hash
Role	ENUM	—	User role: Patient, Doctor, or Admin
Phone	VARCHAR	—	Contact phone number
Created_at	DATETIME	—	Account creation timestamp
Status	BOOLEAN	—	Account status (Active/Inactive)

2. Patient Table

Field Name	Data Type	Key	Description
Patient_id	INT	PK	Unique patient identifier
User_id	INT	FK	References User(user_id)
Age	INT	—	Patient age in years
Gender	VARCHAR	—	Gender (Male / Female / Other)
Height	DECIMAL	—	Height in centimeters
Weight	DECIMAL	—	Weight in kilograms
Blood_pressure	VARCHAR	—	Baseline blood pressure reading
Cholesterol	FLOAT	—	Cholesterol level
Fasting_blood_sugar	FLOAT	—	Fasting blood sugar level
Smoking_status	BOOLEAN	—	Smoking history flag
Family_history	BOOLEAN	—	Family history of heart disease
Created_at	DATETIME	—	Record creation timestamp

3. Medical Assessment Table

Field Name	Data Type	Key	Description
Assessment_id	INT	PK	Unique assessment identifier
Patient_id	INT	FK	References Patient(patient_id)
Chest_pain_type	VARCHAR	—	Type of chest pain experienced
Resting_bp	FLOAT	—	Resting blood pressure (mmHg)
Cholesterol	FLOAT	—	Serum cholesterol (mg/dl)
Fasting_bs	BOOLEAN	—	Fasting blood sugar > 120 mg/dl
Resting_ecg	VARCHAR	—	Resting ECG results
Max_heart_rate	INT	—	Maximum heart rate achieved
Exercise_angina	BOOLEAN	—	Exercise-induced angina status
Oldpeak	FLOAT	—	ST depression induced by exercise
St_slope	VARCHAR	—	Slope of peak exercise ST segment
Vessels	INT	—	Number of major vessels colored by fluoroscopy
Thalassemia	VARCHAR	—	Thalassemia diagnostic result
Assessment_date	DATETIME	—	Date and time of assessment

4. Prediction Table

Field Name	Data Type	Key	Description
Prediction_id	INT	PK	Unique prediction identifier
Assessment_id	INT	FK	References Medical Assessment(assessment_id)
Model_name	VARCHAR	—	ML model used for evaluation
Prediction	VARCHAR	—	Predicted disease outcome
Risk_score	FLOAT	—	Calculated risk percentage (0-100%)
Risk_level	VARCHAR	—	Risk classification (Low, Moderate, High)
Confidence_score	FLOAT	—	Model confidence score
Prediction_date	DATETIME	—	Timestamp of prediction generation

5. Explainable AI Table

Field Name	Data Type	Key	Description
Explanation_id	INT	PK	Unique explanation identifier
Prediction_id	INT	FK	References Prediction(prediction_id)
Method	VARCHAR	—	XAI Method used (SHAP / LIME)
Feature_name	VARCHAR	—	Key medical feature evaluated
Feature_value	VARCHAR	—	Patient's feature value
Contribution	FLOAT	—	Feature contribution score to prediction
Explanation_text	TEXT	—	Human-readable explanation text

6. Recommendation Table

Field Name	Data Type	Key	Description
Recommendation_id	INT	PK	Unique recommendation identifier
Prediction_id	INT	FK	References Prediction(prediction_id)
Diet_advice	TEXT	—	General dietary recommendations
Exercise_advice	TEXT	—	Physical activity and exercise guidelines
Lifestyle_advice	TEXT	—	General lifestyle modifications
Medical_advice	TEXT	—	Clinical recommendations & specialist care guidance
Created_at	DATETIME	—	Timestamp of recommendation creation

7. AI Chat Table

Field Name	Data Type	Key	Description
Chat_id	INT	PK	Unique chat log identifier
User_id	INT	FK	References User(user_id)
Question	TEXT	—	User's query/question text
Ai_response	TEXT	—	AI-generated conversational response
Created_at	DATETIME	—	Chat interaction timestamp

8. Report Table

Field Name	Data Type	Key	Description
Report_id	INT	PK	Unique report identifier
Patient_id	INT	FK	References Patient(patient_id)
Prediction_id	INT	FK	References Prediction(prediction_id)
Report_file	VARCHAR	—	PDF report file path or filename
Generated_by	INT	FK	References User(user_id) who generated report
Generated_at	DATETIME	—	Report generation timestamp

9. ML Model Table

Field Name	Data Type	Key	Description
Model_id	INT	PK	Unique model identifier
Model_name	VARCHAR	—	Model name/identifier
Algorithm	VARCHAR	—	Algorithm (Random Forest, SVM, XGBoost, etc.)
Accuracy	FLOAT	—	Model accuracy score
Precision	FLOAT	—	Model precision score
Recall	FLOAT	—	Model recall score
F1_score	FLOAT	—	F1 score
Auc_score	FLOAT	—	ROC-AUC evaluation score
Version	VARCHAR	—	Model version string
Trained_date	DATETIME	—	Model training completion date
Status	BOOLEAN	—	Model active status flag

10. Admin Table

Field Name	Data Type	Key	Description
Admin_id	INT	PK	Unique administrator identifier
User_id	INT	FK	References User(user_id)
Permissions	TEXT	—	JSON/Text specification of permissions
Last_login	DATETIME	—	Last administrative login timestamp
Status	BOOLEAN	—	Admin account active status